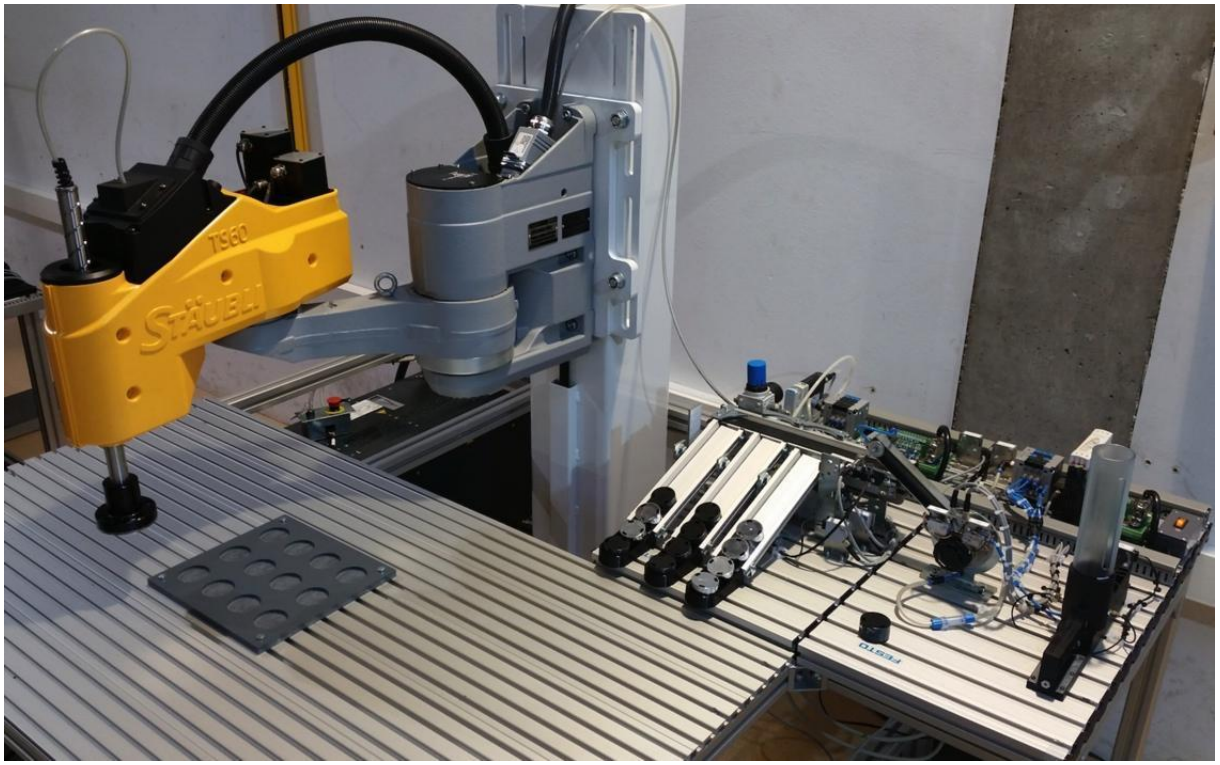
Robot Manipulation

Pick and Place with the Stäubli TS60



This document aims to explain the work to be done in the practice of assembly using the Staubli robot. You can consult the "SRS reference manual" and the "VAL3 reference manual, version 6" when necessary. You need to have completed the SRS "Tutorial", in which you learn how to create an application and simulate it with the emulator. You must have the SRS program installed and use the university license.



IMPORTANT. You should carefully follow the instructions below to avoid any chance of an accident.

- Make sure that in your code the maximum speeds of mdesc type variables are never higher than 30. Never test at a speed higher than 30% of the maximum speed.
- Always lower the speed from the MCP to 10% the first time you try a program. If the program works properly, you can then increase the speed from the MCP to 100% (30% of the maximum speed defined in the code).
- Before pressing the Move / Hold button, locate the STOP and EMERGENCY STOP buttons on the MCP, the external emergency stop switch, and check that the barriers are properly armed.
- Before pressing the Move / Hold button, make sure that no one is in the environment marked by the manipulator's security barriers.
- During execution, use the Move / Hold button to stop and restart execution.
- During execution, use the STOP button to stop the entire application. Be careful, it won't stop immediately, press Move / Hold first.
- Only in case of danger to a person or the manipulator do you press the EMERGENCY STOP buttons.

1. Introduction

This lesson consists in the programming of the Stäubli TS60 to develop a simplified robot manipulation in a Pick and Place environment.

The student will work with the computer and also directly with the robot. Like there is only one robot for all the students, it will be required to work as much as possible in simulation.

The lesson consists of different levels, allowing the students to reach one or another, depending on their skills and interest.

2. Cell description

In this section, the different elements of the cell are presented, see Fig.1.

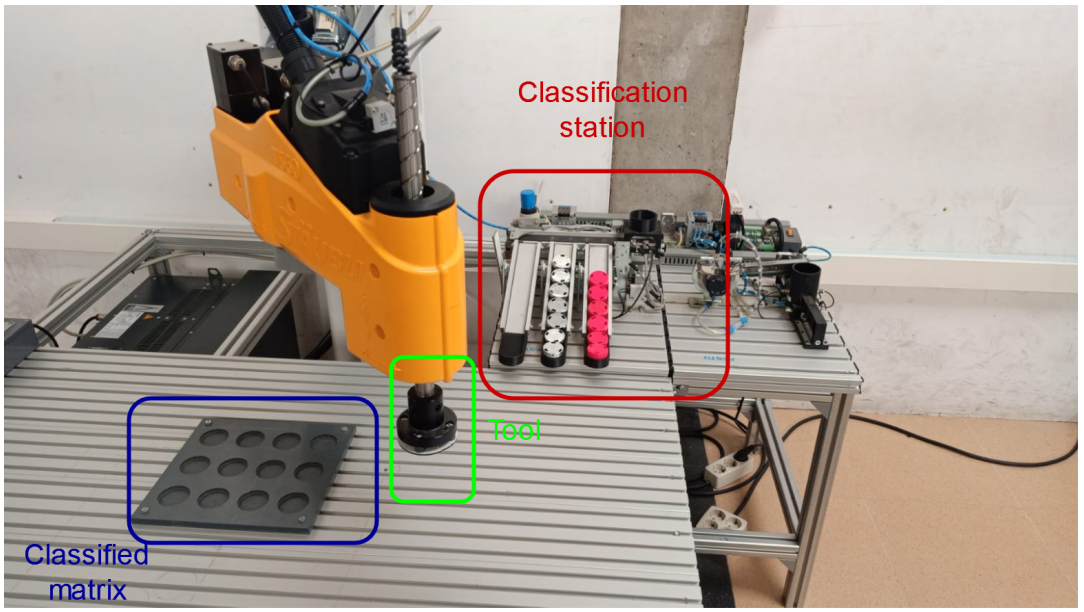


Figure 1: Cell elements

2.1 Parts

In this lesson two parts have to be classified: the *non-metallic cylinder* and the *metallic cylinder* (see Fig.2).



Figure 2: Metallic cylinders and non-metallic cylinders

2.2 Tool

The tool is the mechanical element added to the robot to be able to operate with the different parts. In this case, the tool (see Fig.3) is composed of one suction pad that allows the robot to suction the cylinders. This suction pad is powered by the vacuum center (see Fig.4).



Figure 3: Tool of the robot

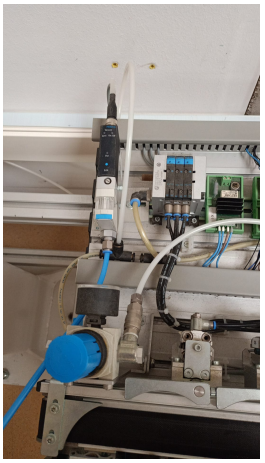


Figure 4: Vacuum center

The tool can be defined as:

x	y	z	rx	ry	rz	gripper	otime	ctime
0	0	20	0	0	0	io:valve1[0]	0.3	0.3

In order to control the vacuum, two outputs are used. The bOut0 controls the generation of the vacuum and the bOut1 controls a blow in order to open the pressure chamber. The sequence to control the tool must be the one represented in Fig.5.

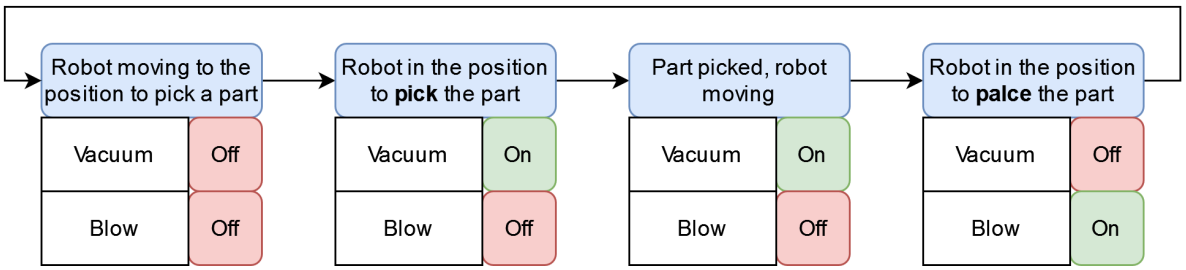


Fig.5: Vacuum control sequence

2.3 Classification station

The *Classification station* is a combination of devices with the purpose of supplying and classifying the cylinders. It consists of different elements, see Fig.6. The ramps, where the parts would be classified, following the ramps names. The trolley that would move the parts from the right part to the left part of the *Classification station*. The *presence sensor 1* that detects if there is a part ready to be supplied. The *presence sensor 2* that detects if there is a part to be classified, in the same position of the trolley but in the opposite lateral, there is the *inductive sensor* that would classify the part. The *barrier* that stops the part, in order to be classified. The *class-actuator 1* and *class-actuator 2* that allows you to put the classified part in the desired ramp. Finally, the *presence sensor 3* is activated when one part goes down in one of the ramps.

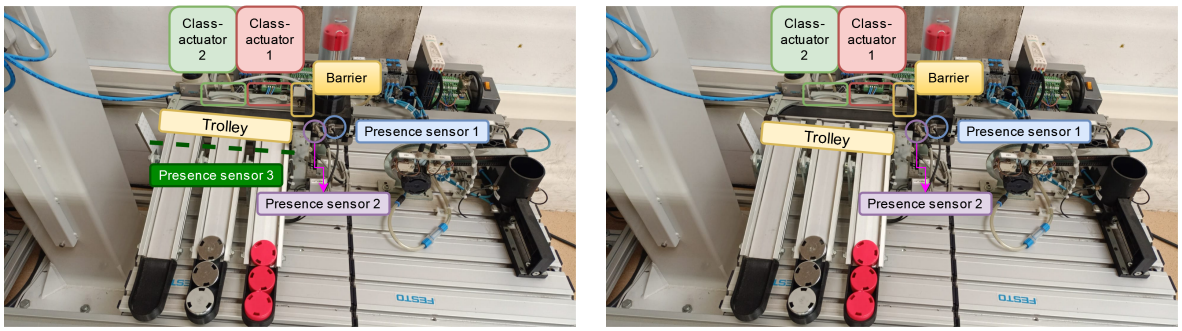
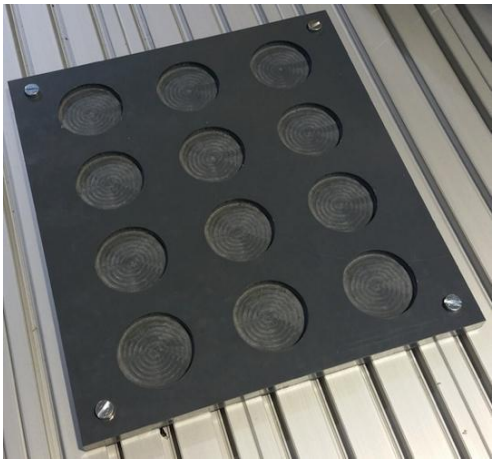


Figure 6: Classification station

2.4 Classified matrix

The *Classified matrix* is a table where there are different positions to place the cylinders, see Fig. 7.



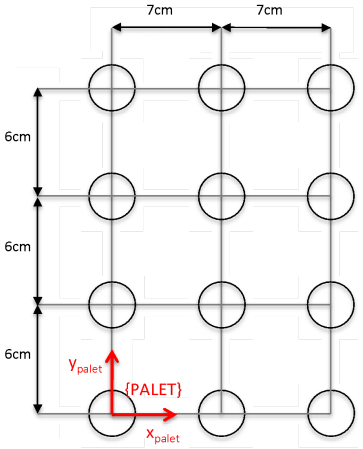


Figure 7: Classified matrix

It is recommended to create a frame related to this object, in order to define the position in an optimal way.

2.5 Inputs/Outputs map

Signal	Logic	Description
bOut0	On (suction) / Off (no suction)	Control of the solenoid valve to open the vacuum chamber in order to create a depression in the suction pad.
bOut1	On (blow) / Off (no blow)	Control of the solenoid valve to activate the blow in the suction pad.
bOut2	On (trolley moving) / Off (trolley stopped)	Control of the *trolley*.
bOut3	On / Off	Control of the actuator of the *Class-actuator 1*.
bOut4	On / Off	Control of the actuator of the *Class-actuator 2*.
bOut5	On / Off	Control of the actuator of the *barrier*.
bIn0	On (presence) / Off (no presence)	Control of the presence of a part sucted in the tool.

bIn1	On (presence) / Off (no presence)	*Presence sensor 2*
bIn2	On (metallic) / Off (no-metallic)	*Inductive sensor*
bIn3	On (presence) / Off (no presence)	*Presence sensor 1*
bIn4	On (presence) / Off (no presence)	*Presence sensor 3*
bIn5	On (extended) / Off (not extended)	Control of the position of the *Class-actuator 1*.
bIn6	On (contracted) / Off (not contracted)	Control of the position of the *Class-actuator 1*.
bIn7	On (extended) / Off (not extended)	Control of the position of the *Class-actuator 2*.
bIn8	On (contracted) / Off (not contracted)	Control of the position of the *Class-actuator 2*.

In order to declare this inputs/outputs in the program, they have to been declared as variables, add these lines to your code:

```
dioLink(pick, io:bOut0)
dioLink(place, io:bOut1)
dioLink(partPresence, io:bIn0)
```

The student would have to add the rest of inputs/outputs in order to be able to develop the Level 3.

2.6 Examples

For a better understanding of the system, in this section, some examples are shown, see Fig. 8, Fig. 9, and Fig. 10.



Figure 8: Part ready and detected (by the *presence sensor 1)* in the beginning of the *Classification station*.

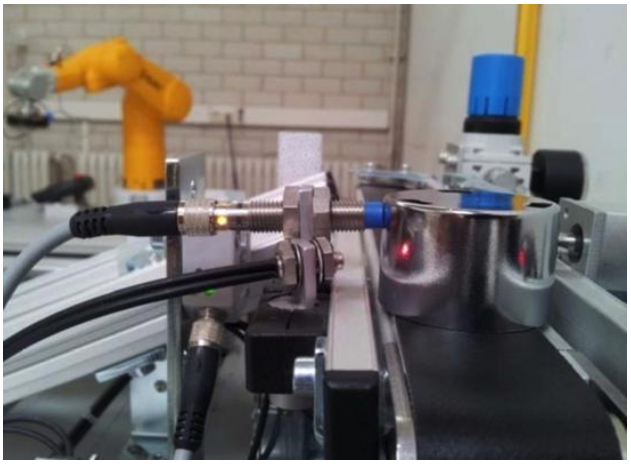


Figure 9: Part stopped in the *barrier* and detected by the *inductive sensor* and the *presence sensor 2*.



Figure 10: Part stopped in the *barrier* and detected by the *inductive sensor* and the *presence sensor 2*, the *Class-actuator 1* is activated in order to classify the part.

3. Memory

The *Pick and place with the Staübli TS60* practical lesson consist of different levels. In each one, the student should program the application on his computer, using SRS. When the application is ready in simulation, the student must show it to the teacher in order to have the acknowledgment to use the robot. When the robot is available and the application is ready, the student can test the program in the real robot.

In the simulation scenario, you have the environment already prepared with the different elements of the laboratory. You can find the file “Entorn_TS60.zip” in the moodle, with the files related to the environment, you must load this file to the SRS.

In the simulation, the student will develop the algorithm required to perform the demands but will put the positions of the robot in a safe place, keeping a conservative distance between the different elements. Would be in the real robot, when the student will adjust the points to the exact required position.

3.1 Level 1: Basic classification

The robot has to classify two parts, a *non-metallic cylinder* and a *metallic cylinder* (remember Fig. 2. The student will manually put one part in each ramp, see Fig. 11, and program the robot in order to place each one in the *Classified matrix*, see Fig. 12. The robot should pick (remember the vacuum control sequence, Fig. 5) one part from each classification ramp (the *dismiss ramp* is not required) and place it in the *Classified matrix*.



Figure 11: Level 1, ramp configuration

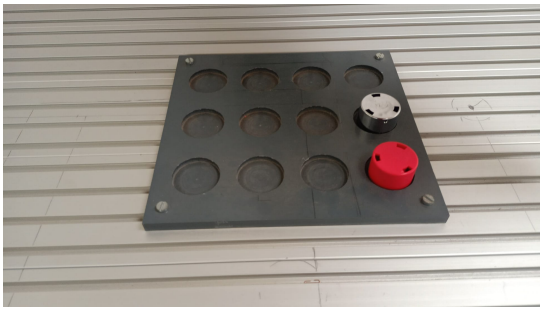


Figure 12: Level 1, *Classified matrix* final configuration

3.2 Level 2: Continuous feeding

The robot has to classify the two parts (*non-metallic cylinder* and *metallic cylinder*) in a continuous way. The student will manually put four parts in each ramp and program the robot to place each one in the *Classified matrix*.

3.3 Level 3: Continuous feeding with automated classification

In this Level the robot would have to take control of the different devices of the cell, in order to make an automated classification. The student will manually put an aleatory sequence of parts in the *Classification station* source tower (see Fig.13), and the application must create a configuration of the *Classified matrix* like the one presented in Fig. 14.

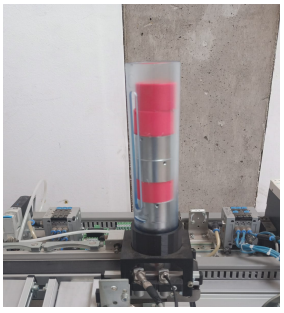


Figure 13: Example of an aleatory sequence of parts in the source tower of the *Classification station*



Figure 14: Level 3 configuration of the *Classified matrix*

4. Evaluation

The students have to present a report explaining the practical lesson and how they develop the code in order to face the different problems. They have to present also a video of each performance of the robot developed.

The evaluation criteria will follow different points:

- Level reached.
- Performance of the robot in each level.
- Use of tools and frames.
- Intuitive messages in the MCP.
- Good structuration of the code, self-explaining code, and usage of the optimal functions.
- Good presentation of the report.